

GROMOS Peptide Workflow: Troubleshooting and Installation

Purpose

This document combines three things:

- a troubleshooting log for the errors encountered while building and running the peptide system,
- a practical installation guide for GROMOS++ and GROMOSXX,

Troubleshooting Log and Fixes

1. Missing hydrogens

Symptom: The coordinates were incomplete or atoms were placed at the origin.

Fix: Run `gch` to complete the structure and generate `peptide_fixed_h.cnf`.

Meaning: hydrogen completion.

2. Wrong coordinate count in restraint files

Symptom: “not enough coordinates in REFPOSITION block”

Fix: Rebuild `.rpr` from the full solvated coordinate file and keep all atoms; only rename the block from `POSITION` to `REFPOSITION`.

Meaning: the restraint reference file must match the full system.

3. FORCE block mismatch

Symptom: “last energy group has to end with last atom”

Fix: Make the last atom number in the `FORCE` block equal the final atom of the current coordinate file.

Meaning: energy groups must match the actual system size.

4. SYSTEM block placeholder

Symptom: “Read NSM for NSM”

Fix: Replace the placeholder by the real number of solvent molecules.

Meaning: the `SYSTEM` block must reflect the actual box contents.

5. MULTIBATH placeholder

Symptom: “Read LSTATM for LAST”

Fix: Replace `LSTATM` with the last atom index of the full system.

Meaning: temperature baths require a real atom index.

6. Wrong solvent count

Symptom: The number of positions in the coordinate file did not match the topology and the SYSTEM block.

Fix: Recalculate the number of water molecules and update SYSTEM and the FORCE block accordingly.

Installation Manual for GROMOS++ and GROMOSXX

Prerequisites

You usually need:

- a C/C++ compiler (gcc or clang),
- a build system (cmake or make, depending on the distribution),
- standard development tools,
- optional linear algebra libraries if requested by your build configuration.

General installation principle

The exact build system can vary depending on the distribution you downloaded from the GROMOS project. The safest approach is:

1. unpack the source archive,
2. read the project-specific README / INSTALL notes,
3. configure the build directory,
4. compile,
5. add the resulting binaries to your PATH,
6. verify the commands with `which`.

Suggested build outline

GROMOS++

```
cd gromos++
mkdir build
cd build
cmake ..
make -j4
```

GROMOSXX

```
cd gromosXX
mkdir build
cd build
cmake ..
make -j4
```

Add the executables to PATH

Add the directories containing the binaries to your shell configuration file (for example `.zshrc` or `.bashrc`):

```
export PATH=/path/to/gromos++/bin:$PATH
export PATH=/path/to/gromosXX/bin:$PATH
```

Verify the installation

Check that the tools can be found:

```
which make_top
which com_top
which check_top
which pdb2g96
which gch
which sim_box
which ion
which mk_script
which md
which frameout
```